



Luigi Capogrosso

POSTDOCTORAL RESEARCHER AT THE INTERDISCIPLINARY TRANSFORMATION UNIVERSITY OF AUSTRIA (IT:U)

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Education

Postdoctoral Researcher

INTERDISCIPLINARY TRANSFORMATION UNIVERSITY OF AUSTRIA (IT:U)

Linz, Austria

Aug. 2025 - Present

- Research interests: Efficient Machine Learning, Learning-Enabled Cyber-Physical Systems, and Representation Learning.

Ph.D. in Artificial Intelligence (*Cum Laude*)

POLYTECHNIC OF TURIN, IN COLLABORATION WITH THE UNIVERSITY OF VERONA

Verona, Italy

Nov. 2021 - Nov. 2024

- Thesis title: "Learning-Based Methods for Enabling On-Edge, Accurate, Sustainable, and Human-Centered Intelligent Manufacturing".

M.Sc. in Computer Engineering for Robotics and Smart Industry (*Cum Laude*)

UNIVERSITY OF VERONA

Verona, Italy

2019 - 2021

- Thesis title: "Embedded Deep Learning for Computer Vision: A Case Study on Object Tracking".

B.Sc. in Computer Science

UNIVERSITY OF VERONA

Verona, Italy

2016 - 2019

- Thesis title: "Linux Kernel: Organization, Structures and Process Management".

Teaching Experiences

Lecturer

INTERDISCIPLINARY TRANSFORMATION UNIVERSITY OF AUSTRIA

Linz, Austria

2025 - 2026

- Course "Physical AI in the Space Domain (2025/2026)" of the Master's degree in Science Interdisciplinary Computing.

Teaching Assistant

UNIVERSITY OF VERONA

Verona, Italy

2020 - 2025

- Course "Topics on Computer Architecture and Operating Systems (2024/2025)" of the Bachelor's degree in Computer Science.
- Course "Hardware Components Design on FPGA (2023/2025)" of the Bachelor's degree in Computer Science.
- Course "Systems Design Laboratory (2023/2024)" of the Master's degree in Computer Engineering for Robotics and Smart Industry.
- Course "Machine Learning & Artificial Intelligence (2022/2023)" of the Master's degree in Computer Engineering for Robotics and Smart Industry.
- Course "Computer Vision & Deep Learning (2022/2023)" of the Master's degree in Artificial Intelligence.
- Course "Signal and Image Processing (2022/2023)" of the Bachelor's degree in Computer Science.
- Course "Topics of Operating Systems and Networks (2020/2021)" of the Bachelor's degree in Bioinformatics.
- Course "Network Programming and Security (2020/2021)" of the Bachelor's degree in Computer Science.
- Course "Programming I (2020/2021)" of the Bachelor's degree in Computer Science.

Reviewer Experiences

JOURNALS

2026	IEEE Sensors Journal , IEEE Inc.
2023 - Present	International Journal of Computer Vision , Springer Netherlands.
2023 - Present	IEEE Internet of Things Journal , IEEE Inc.
2023 - Present	Future Generation Computer Systems , Elsevier B.V.
2022 - Present	Pattern Recognition , Elsevier Ltd.

CONFERENCES

2026	Conference on Neural Information Processing Systems (NeurIPS) , OpenReview
2024 - Present	Computer Vision and Pattern Recognition Conference (CVPR) , CVF/IEEE Inc.
2023 - Present	International Conference on Learning Representations (ICLR) , OpenReview.
2023 - Present	European Conference on Computer Vision (ECCV) , Springer.

Industry Experiences

Research Fellow

CONFINDUSTRIA VERONA

Verona, Italy
Dec. 2024 - Jul. 2025

- Grant title: "Sustainability: Tools and Models for Innovation Towards Carbon Neutrality". FSE Project Code: 1001-0001-231- 2024 (DGR No. 231 of 13/03/2024). Research and development of innovative strategies for achieving carbon neutrality.

Research Assistant

HUMATICS | SYS-DAT GROUP

Verona, Italy
Jan. 2021 - Aug. 2021

- Deep learning models are often computationally expensive, hindering their use in resource-constrained edge systems. During this period, my work was focused on implementing pruning and quantization optimization techniques on company confidential deep learning models.

Research Fellow

EDALAB S.R.L.

San Giovanni Lupatoto, Italy
Oct. 2019 - Dec. 2021

- Communication protocols are the foundation of Internet of Things (IoT) applications. During this period, my work was focused on implementing the Modbus, OPC-UA, and MQTT protocols inside the company's BOX-IO embedded system.

Committees

2027	Technical Program Committee , Design, Automation and Test in Europe Conference (DATE).	Dresden, Germany
2026	Special Session Organizer , IEEE Sensor Applications Symposium (IEEE SAS).	Vitória, Brazil
2025	Area Chair , 23rd International Conference on Image Analysis and Processing (ICIAP).	Rome, Italy
2025	Ph.D. Forum Chair , 3rd International Conference on Frontiers of Artificial Intelligence (FAIEMA).	Stavanger, Norway
2025	Program Committee , International Conference on Omni-Layer Intelligent Systems (COINS).	Madison, U.S.A.
2024	Program Committee , Unifying Representations in Neural Models Workshop (UniReps) at NeurIPS.	Vancouver, Canada

Honors & Awards

2026	Outstanding Reviewer , 19th European Conference on Computer Vision (ECCV).	Malmö, Sweden
2026	Best Paper Award , IEEE International Workshop on Metrology for Automotive.	Brescia, Italy
2026	Outstanding Dissertations Award , European Design and Automation Association (EDAA).	Verona, Italy
2025	Outstanding Reviewer , IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR).	Nashville, U.S.A
2022	Honorable Mention , ACM/IEEE ICCAD TinyML Design Contest.	San Diego, U.S.A
2016 - 2021	Student Excellence Grant , University of Verona.	Verona, Italy

Personal Qualities & Interests

Scientific interests	Efficient Deep Learning, Representation Learning, Cyber-Physical Systems, Industry 4.0, Space Exploration.
Programming	Python, PyTorch, C, C++.
Languages	Italian, English (Certified Level B2).

Selected Publications

Performance Analysis of Edge and In-Sensor AI Processors: A Comparative Review

L CAPOGROSSO , P BONAZZI, M MAGNO	2026
• International Instrumentation and Measurement Technology Conference (I2MTC).	

A Comprehensive Survey on Deep Learning-based Predictive Maintenance

U KHAN , D CHENG, F SETTI, F FUMMI, M CRISTANI, L CAPOGROSSO	2026
• ACM Transactions on Embedded Computing Systems.	

LO-SC: Local-only Split Computing for Accurate Deep Learning on Edge Devices

L CAPOGROSSO , E FRACCAROLI, M CRISTANI, F FUMMI, S CHAKRABORTY	2025
• International Conference on VLSI Design (VLSID).	

MTL-Split: Multi-Task Learning for Edge Devices using Split Computing

L CAPOGROSSO , E FRACCAROLI, S CHAKRABORTY, F FUMMI, M CRISTANI	2024
• Design Automation Conference (DAC).	

A Machine Learning-oriented Survey on Tiny Machine Learning

L CAPOGROSSO , F CUNICO, DS CHENG, F FUMMI, M CRISTANI	2024
• IEEE Access.	